MIL-C-45200(Ord)
1 December 1958
SUPERSEDING
FA-PD-MI-2260
21 May 1957

MILITARY SPECIFICATION

CARTRIDGE, BALL, CALIBER .22, M65 (HORNET)

1. SCOPE

1.1 This specification covers Cartridge, Ball, Caliber .22, M65 (Hornet) intended for use in Caliber .22, survival weapons.

2. APPLICABLE DOCUMENTS

2.1 The latest revisions of the following specifications, standards, drawings and publications, unless otherwise specified, of the issue in effect on the date of invitation for bids, form a part of this specification:

SPECIFICATION

MILITARY

MIL-A-625 - Ammunition, Small Arms, General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Inspection Terms and Definitions
MIL-STD-120 - Marking for Shipment and Storage

DRAWINGS

ORDNANCE CORPS

C-7553664 - Cartridge, Ball, Caliber .22, M65 (Hornet)
B-7643674 - Classification of Cartridge Case Defects
F-8594079 - Packing and Marking for Cartridges, Ball, Caliber .22, M65 (Hornet) in Carton, in Waterproof Bag

FSC 1305

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Frankford Arsenal

GL-7553664 - List of Inspection Gages for Assembly

PUBLICATIONS

Department of the Army

SM-9-5-1305 - Department of the Army Supply Manual, Ammunition, through 30MM

ORDNANCE CORPS

ORD-SIP-S315 - Visual Inspection Standards for Small Arms Ammunition
ORD-M608-PM - Manual of Test Methods for Small Arms Ammunition Vol. III

(Copies of Specification, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other Publications.-The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification, Ratings, Rules and Regulations.

(Copies may be purchased from the Uniform Classification Committee, 202 Chicago Union Station, Chicago 6, Illinois.)

AMERICAN TRUCKING ASSOCIATION

National Motor Freight Classification, Uniform Ratings, Rules and Regulations.

(Copies may be purchased from the National Classification Board, 1424 Sixteenth Street, N.W., Washington 6, D. C.)

INTERSTATE COMMERCE COMMISSION - 49 CFR 71-78

Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Interstate Commerce Commission regulations are now a part of the Code of Federal Regulations available from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Orders for the above Publication should cite "49 CFR 71-78".)

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3. REQUIREMENTS

3.1 General.-The cartridge shall comply with the requirements specified on the applicable drawings, referenced specifications and the following:
3.2 Preproduction sample.-Unless otherwise directed by the contracting officer, a preproduction sample shall be required.
3.3 Bullet Extraction.-The cartridge shall withstand a minimum bullet pull of 25 pounds when tested in accordance with 4.4.1.
3.4 Mercurous Nitrate.-The cartridge case shall not show splits when subjected to the mercurous nitrate test referenced in 4.4.2.
3.5 Velocity.-The average velocity of the cartridges shall be 2750 feet per second (ft/sec) ± 50 ft/sec at 15 feet from the muzzle when tested in accordance with 4.4.4.
3.6 Waterproof .-The average velocity of the cartridges after submersion in water shall not vary from the average velocity of the same lot by more than 100 feet per second when tested in accordance with 4.4.5.
3.7 Chamber Pressure.-The average pressure of the cartridges shall not exceed 43,000 pounds per square inch when tested in accordance with 4.4.6.
3.8 Accuracy.-The average extreme spread of all targets at 100 yards shall not exceed 2.5 inches when tested in accordance with 4.4.7.
3.9 Function and Casualty.-The cartridge shall function without casualty at ambient temperatures, at -40° to -45° Fahrenheit (f) and at ± 125° to ± 130°F when tested in accordance with 4.4.8.

4. QUALITY ASSURANCE PROVISIONS

4.1 General.-The quality assurance provisions in this section and other documents referenced in this specification are applicable to inspection performed by or for the Government. Reference shall be made to Standard MIL-STD-109 for definition of inspection terms used. Inspection shall be in accordance with Specification MIL-A-625 except as specified herein.
4.1.1 Contractor inspection.-Unless otherwise specified herein, the inspections required by this section and other documents referenced in this specification shall be performed by the contractor.
4.1.2 Contractor Quality Assurance System.-The contractor shall provide and maintain an adequate quality assurance system, acceptable to the Government, covering the supplies under this specification. Results of all examinations and tests performed under the quality assurance system shall be made available to the Government. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The written description will be considered acceptable when, as a minimum, it provides the quality assurance required by this specification and other referenced documents. The contractor shall notify the Government of and obtain approval for any change to the written procedure that affects the degree of assurance required by this specification or other referenced documents.

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4.1.3 Government Verification.-All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of monitoring the operations to determine whether the practices, methods and procedures of the written inspection plan, as provided in the quality assurancy system, are being properly applied. Verification may also consist of Government product inspection to measure the quality of the product offered for acceptance. Deviations from the prescribed or agreed upon procedures or instances of poor practices which might have a detrimental effect upon the quality of the product will be immediately called to the attention of the contractor. Failure of the contractor to correct deficiencies promptly will be cause for suspension of acceptance until correction has been made or until conformance of the product to the prescribed criteria has been demonstrated. To avoid interference with operations, the contractor shall designate a responsible official or officials to whom the Government inspector will report such instances.

4.2 Preproduction Inspection.

4.2.1 Preproduction Sample.-Prior to the start of regular production, the contractor shall submit a preproduction sample to a Government approved facility for evaluation. The sample shall consist of 2,000 cartridges from the contractor's initial production, manufactured in the same manner, using the same materials, equipment, processes and procedures as will be used in regular production. The contracting officer will furnish instructions for shipment of the sample to a designated facility for examination and tests. Any production under the contract prior to approval of the preproduction sample shall be at the risk of the contractor. After the start of production, additional preproduction samples shall be submitted for approval immediately subsequent to each major change in material or manufacturing processes as determined by the contracting officer.

4.2.2 Inspection.-The preproduction sample shall be examined and tested for all requirements of the drawings and specifications including all those characteristics listed in 4.3.3.4 and 4.3.4. All testing shall be conducted in accordance with 4.4. The Government reserves the right to require new preproduction samples until such time as an acceptable sample is submitted.

4.3 Inspection Provisions.-Alternate inspection procedures and inspection equipment may be used by the contractor when such procedures and equipment provide, as a minimum, quality assurance required in the contractual documents. The inspection procedures or equipment specified in this specification and referenced documents will prevail in case of dispute. However, in all ballistic testing, the use of the equipment specified and the methods of test prescribed in the specification and other documents referenced herein shall be mandatory.

4.3.1 Material.-Reports of objective tests performed on the material by the contractor shall be required to insure compliance with the chemical and physical requirements. Reports shall include positive identification of the material and part, the applicable requirements, the specific test

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performed and the results obtained. Numerical results of each specimen or sample inspected shall be included except that where numerical results are impossible an adjective description shall be used. Initial test reports shall accompany the preproduction sample. Prior to inspection, the inspector shall determine from the information stated on the test reports whether the material submitted complies with requirements. Components made from material which does not comply shall be rejected.

4.3.2 Lot.

4.3.2.1 Submittal of Product.-The contractor shall submit the product in accordance with Standard MIL-STD-105.

4.3.2.2 Size of Lot.-Unless otherwise specified by the procuring agency, an ammunition lot shall consist of the number of cartridges produced in one day. However, where a day's production is relatively small, an ammunition lot may include the production of two or more consecutive days but not more than 2,000,000 cartridges.

4.3.2.3 Lot Identification.-Each lot of ammunition shall be identified in accordance with Specification MIL-A-625.

4.3.3 Examination.-Examination shall be performed as indicated below:

4.3.3.1 Critical Defects.-Each cartridge in the lot shall be visually examined and gaged for critical defects. All cartridges containing such defects shall be removed from the lot.

4.3.3.2 Sampling.-Samples for examination shall be selected in accordance with Inspection Level II of Standard MIL-STD-105. Each cartridge of the sample shall be visually examined and gaged for all major and minor defects listed in Table I. If the number of defective cartridges is equal to or greater than the applicable rejection number, the lot shall be rejected subject to reinspection in accordance with Specification MIL-A-625.

4.3.3.3 Acceptable Quality Levels.-The acceptable quality levels with respect to visual and dimensional defects shall be as follows:

DEFECTS	PERCENT
Major	.25
Minor	1.5

4.3.3.4 Classification of Defects.-The visual and dimensional defects shall be classified as specified in Table I.

TABLE I

No.	Defect	Critical	Major	Minor	Major or Minor
	Visual 1/				
	Cartridges				
1	Discolored, dirty, oily smeared			X	
2	Corroded		X		
3	Mixed ammunition commercial type hornet cartridge (partially jacketed bullet)	X			

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TABLE I (Contd.)

No.	Defect	Critical	Major	Minor	Major or Minor
	Visual 1/				
	Cartridges (Contd)				
4	Crooked			X	
	Case				
5	Dent				X
6	Split case 2/		X		
7	Perforated case 2/		X		
8	Draw scratch				X
9	Scratch			X	
12	Scaly metal				X
14	Fold			X	
15	Wrinkle			X	
16	Buckle			X	
17	Bulge			X	
18	Illegible or missing head stamp			X	
19	Defective head			X	
20	Defective mouth			X	
	Bullet				
22	Dent			X	
23	Scratch			X	
24	Split bullet jacket		X		
25	Loose bullet		X		
27	Scaly metal (bullet)				X
28	Upset (crooked) point			X	
29	Exposed steel (clad jacket)			X	
30	Blunt point			X	
	Primer				
32	No primer	X			
33	Cocked primer	X			
34	Inverted primer	X			
35	Loose primer		X		
36	Nicked or dented primer			X	
37	No waterproofing (primer pocket joint)			X	
	Gaging				
	Cartridge				
38	Long overall length			X	
39	Short overall length			X	
40	Cartridge profile failure (requiring more than 10 pounds dead weight to insert in profile and alignment gage)		X		
	Case				
41	Large diameter head		X		
42	Small diameter head			X	
43	Thick head		X		
44	Thin head			X	
45	(Primer) High primer		X		
46	Low Primer		X		

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1/ refer to ORD-SIP-S315, Caliber .30 Section for definite classification of defects 1 through 37, with the exception of defect 4.

2/ To be counted as a major defect if no loss of propellant occurs.
To be counted as a critical defect if loss of propellant occurs.

4.3.4 Tests.-The tests listed in Table II shall be conducted in accordance with the methods and procedures specified in 4.4.
4.3.4.1 Test Samples.-The quantities for the various tests shall be as specified in Table II. Only cartridges having met the visual and dimensional requirements shall be used in the tests, and shall have been selected in such a manner that the sample is representative of the entire lot. The cartridges selected shall be thoroughly mixed before dividing into small samples for the various tests.

TABLE II

Test	No. of Cartridges Temperature			Requirement Paragraph
	Ambient	-40°F	+125°F	
Bullet extraction 1/	50			3.3
Mercurous Nitrate 1/	20			3.4
Velocity 2/	10			3.5
Waterproof 2/	20			3.6
Chamber pressure 2/	10			3.7
Accuracy 2/	50			3.8
Function and Casualty3/				3.9
Rifle-Shotgun, Survival Caliber .22/ .410, M6	100	76	76	
Rifle, Survival,Caliber .22, M4	100	76	76	

1/ Failure of any cartridge to comply with the applicable requirement shall be cause for rejection of the lot. Retest may be conducted with double the test sample, provided that no more than one defective cartridge is found in the initial test. If any additional defective cartridges are found upon retest, the lot shall be rejected.

2/ Failure of the cartridges to comply with the applicable requirement shall be cause for rejection of the lot. Retest may be conducted in accordance with Specification MIL-A-625 with double the quantity used in the initial test. Failure of the cartridges in the retest sample to comply with the applicable requirement shall be cause for rejection of the lot.

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3/ The lot shall be rejected when function and casualty defects plus all other firing test defects exceed the number permitted in Table III. Retest may be conducted in accordance with 4.3.3.3 provided that the number of defects found in the initial test does not exceed the number permitted for the cumulative sample (test plus retest). If the total number of defects in the test and retest exceeds the number permitted for the cumulative sample, the lot shall be rejected.

4.3.4.2 Firing Defects.-Firing defects and number permitted in a sample shall be as specified in Table III.

TABLE III

Defects	No. of Defects Permitted	
	Initial Test	Cumulative (Test plus Retest)
Misfire	1	3
Bullet remaining in bore	0	-1/
Primer leak		
a. Perforation in firing pin indent in primer cup	2	5
b. Escape of gas through primer cup other than "a".	6	16
c. Escape of gas around primer cup 2/	29	80
d. Loose primer	3	8
e. Blown primer 3/	0	1
Case Casualties		
a. Longitudinal split 4/		
(1) Neck and shoulder (L and S)	15	40
(2) Body (J)	3	8
(3) Body (K)	0	1
(4) To head (L)	0	1
(5) Through head (M)	0	1
b. Circumferential Rupture 4/		
(1) Partial shoulder or body (S or J)	3	7
(2) Partial body (K)	0	1
(3) Partial head (L)	0	1
(4) Complete	0	1
Failure to extract 5/	1	3
Bullet stripping before striking terminal or target	0	1

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1/ No retest permitted. Lot shall be rejected.

2/ Gas escapes around more than 50 percent of periphery of cup.

3/ Primer not in pocket after firing and both head of cartridge case and pocket enlarged and distorted (determine visually).

4/ For location of defects indicated by letters in parentheses see Drawing B-7643674.

5/ When failure to extract occurs in Rifle, Survival, Caliber .22/.410, M6, the cartridge case which fails to extract shall be removed from the chamber of the weapon and examined by the inspector. If this examination reveals no evidence of a cartridge defect, then the failure to extract shall be attributed to a malfunction of the extraction system of the weapon, and shall not be counted against the ammunition. The cartridge case will be considered to have extracted normally.

4.3.4.3 Retests.-Retests, when permissible, will be conducted at the request of the contractor. Retest for failure due to excessive firing defects shall be made by firing double the quantities specified in Table II under function and Casualty Test in all the service weapons specified therefore. This procedure for retest shall apply regardless of the weapon or weapons in which the firing defects occurred during the initial tests.

4.3.5 Packaging, packing and marking examination.

4.3.5.1 Sampling.-Samples shall be selected in accordance with Inspection Level I of Standard MIL-STD-105.

4.3.5.2 Acceptable Quality Levels.-The acceptable quality levels with respect to packaging, packing and marking shall be as follows:

	MAJOR (Percent)	MINOR (Percent)
Carton	-	1.5
Waterproof bag	.65	2.5
Box	1.0	2.5

4.3.5.3 Classification of Defects.-Packaging, packing and marking defects shall be as specified in Table IV. (see 5.)

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TABLE IV

No.	Defect	Major	Minor
	Carton		
1.	Carton or filler not as specified on applicable drawing		X
2.	Improper number or positioning of cartridges		X
3.	Torn or ripped carton		X
	Waterproof Bag		
4.	Torn, ripped or improperly sealed waterproof bag	X	
5.	Incorrect or illegible identification of waterproof bag contents (type, caliber, etc.)		X
6.	Incorrect, illegible or missing ammunition lot number on waterproof bag		X
	Box		
7.	FSN missing, incorrect or illegible	X	
8.	DOD Code incorrect	X	
9.	DOD Code missing or illegible		X
10.	Lot number missing or illegible	X	
11.	Marking other than FSN, DOD Code and lot number incorrect, missing or illegible		X
12.	Marking touched-up or repaired with non-permissible material (crayon, chalk, etc.)		X
13.	Protruding nail point on inside of box	X	

4.3.6 Inspection Equipment.-The examination and tests shall be made using equipment listed on GL-7553664, except as provided in 4.3. The Government will determine by means of its monitoring program, that the contractor has available and correctly utilizes, gaging, measuring, and test equipment of required accuracy and precision and that the instruments are of proper type and range to make measurements of the required accuracy. The contractor shall have available a set of master gages, standards and other appropriate instruments for regularly scheduled calibration of his inspection equipment. Records of such regularly scheduled calibration shall be maintained by the contractor and be made available to the Government for review. The calibration of gages, standards and instruments will be periodically checked by authorized Government personnel.

4.4 Test Methods and Procedures.

4.4.1 Bullet extraction.-The cartridges shall be tested in an approved bullet extractor machine. The rate of travel of the test head shall not be less than three nor more than six inches per minute.

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4.4.2 Mercurous Nitrate.-Tests shall be conducted in accordance with publication ORD-M608-PM, Vol. III.

4.4.3 Headspace.-The headspace for test weapons and field service weapons shall conform to the applicable drawing of the weapon. Methods cited in Publication ORD-M608-PM, Vol. III shall be used for this determination.

4.4.4 Velocity.-The test shall be conducted in accordance with Publication ORD-M608-PM, Vol. III.

4.4.5 Waterproof.-The test shall be conducted in accordance with Publication ORD-M608-PM, Vol. III.

4.4.6 Chamber pressure.-The test shall be conducted in accordance with Publication ORD-M608-PM, Vol. III. The oiled obturating cups are to be changed after each shot.

4.4.7 Accuracy.-Five targets of ten shots each shall be fired. The test shall be conducted in accordance with Publication ORD-M608-PM, Vol. III.

4.4.8 Function and Casualty.-In all tests, the weapon shall be at ambient temperature before firing. The tests shall be conducted in accordance with Publication ORD-M608-PM, Vol. III. (See 6.2)

4.4.8.1 Low Temperature Test.-The cartridges shall be cooled to a temperature of -40°F to -45°F and maintained thereat for not less than four hours and fired at that temperature. The ammunition for the low temperature test shall be placed in a controlled-temperature box in such a manner that all cartridges are subjected to a uniform temperature, cooled to a temperature of -40°F to -45°F and maintained thereat for not less than four hours immediately before firing. The cartridges for the M4 weapon shall be in magazines. The controlled-temperature box shall be of sufficient capacity to permit cooling of ammunition to the desired temperature and also to allow free circulation of air. The ammunition shall be fired immediately upon removal from the cold box in the specified weapons, which shall be at ambient temperature. If the weapon is not convenient to the controlled-temperature box the ammunition shall be placed in an insulated chest which has been cooled to a temperature of -40°F to -45°F, transported to the weapon, and fired immediately upon removal.

4.4.8.2 High Temperature Test.-The cartridges shall be heated to a temperature of 125°F to 130°F and maintained thereat for not less than four hours and fired at that temperature. The ammunition for this test shall be stored, conditioned and fired in the same manner as that prescribed for the low temperature test except that the temperature at which the ammunition is conditioned shall be 125°F to 130°F.

4.4.9 Defect Penalty.-In any ballistic test, except function and casualty, in which the occurrence of a firing defect prevents the obtaining of a reliable result for the characteristic being tested, an additional shot shall be fired. That particular test shall not be penalized, but the acceptance sample shall be penalized for such defects in accordance with Table III.

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5. PREPARATION FOR DELIVERY

5.1 Packaging.-Cartridges shall be packaged in accordance with Drawing No. F-8594079 or as requested by the contracting officer.

5.2 Packing.-Unless otherwise specified the cartridges, packaged in waterproof bags, shall be packed in commercial containers.

5.3 Marking and Labeling.-Each waterproof bag shall be labeled in accordance with Drawing No. F-8594079. Each packing container shall be marked in accordance with MIL-STD-129. The markings shall include the descriptive nomenclature and number of cartridges, ammunition lot number, manufacturer's initials, class number, Federal Stock Number, Federal Item Identification Number, Department of Defense Ammunition Code, average weight and cubic displacement.

5.4 Delivery.-A lot, which has been officially inspected and accepted by the Government inspector, shall be considered as delivered when the cartridges are packed, the ammunition boxes sealed and marked as required and the packed ammunition accepted. The lot shall then be shipped or stored in accordance with the terms of the contract or as directed by the contracting officer. When the contract provides for delivery f.o.b. cars at the place of manufacture, the ammunition shall be placed and properly secured in accordance with common carrier regulations as listed in 2.2.

6. NOTES

6.1 Ordering Data.-Procurement documents should specify the title, number and date of this specification.

6.2 Safety precautions should be used in firing high and low temperature tests since the possibility of weapon failure is greatly increased. The gunner should fire by remote control from behind a barricade.

Notice: This specification, together with specifications and drawings pertaining to it and bearing a "Notice" of similar restrictions, is intended for use only in connection with procurement by the United States Government and shall not be used for any other purpose except when specifically authorized by the Chief of Ordnance.

Custodian:
Army - Ord Corps

Preparing Activity:
Ordnance Corps

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